

# Semiconductors

1. Calculate the conductivity of an intrinsic Ge sample with the intrinsic carrier density  $3.2 \times 10^{19}/\text{m}^3$ . Given, mobility of electrons and holes as  $0.36 \text{ m}^2/\text{V-sec.}$  and  $0.17 \text{ m}^2/\text{V-sec.}$  respectively at room temperature.

(Ans:  $2.71 \text{ mho/m}$ )

2. For intrinsic Si, the electrical conductivity is  $4 \times 10^{-4} \text{ mho/m}$  at room temperature. Calculate the electron and hole concentration at room temperature. Given, the mobility of electrons and holes are  $0.14 \text{ m}^2/\text{V-sec.}$  and  $0.040 \text{ m}^2/\text{V-sec.}$  respectively.

(Ans:  $1.388 \times 10^{16}/\text{m}^3$ )

3. Calculate the conductivity of a Ge specimen if the donor impurity added to Ge is  $1.2 \times 10^{23} \text{ atoms/m}$ . Given, mobility of electron is  $3800 \text{ cm}^2/\text{V-sec.}$

(Ans:  $7296 \text{ mho/m}$ )

4. In intrinsic Ge the carrier concentration is  $2.5 \times 10^{19}/\text{m}^3$ . The electron and hole mobilities are  $0.39 \text{ m}^2/\text{V-sec.}$  and  $0.17 \text{ m}^2/\text{V-sec.}$  respectively. Find the resistance of a Ge rod of  $2 \text{ cm} \times 1 \text{ mm} \times 1 \text{ mm}$  dimension.

(Ans:  $8.928 \times 10^3 \text{ ohm}$ )

5. The resistivity of intrinsic InSb at room temperature is  $2 \times 10^{-4} \text{ ohm-cm}$ . If the mobility of electron is  $6 \text{ m}^2/\text{V-sec.}$  and mobility of hole is  $0.2 \text{ m}^2/\text{V-sec.}$  Calculate its intrinsic carrier density.

(Ans:  $5.04 \times 10^{23} / \text{m}^3$ )

6. Calculate the conductivity of a Germanium sample if a donor impurity atom is added  $10^6$  Ge atoms at room temperature. Assume that only electron of each atom takes part in conduction process.

Given, Avogadro's No =  $6.023 \times 10^{23}$  atoms/gm-mole

Atomic weight of Ge = 72.6

Mobility of electrons =  $3800 \text{ cm}^2/\text{V-sec}$

Density of Ge =  $5.32 \text{ gm/cm}^3$ .

(Ans: 2683 mho/m)

7. Evaluate the Fermi function for energy  $kT$  above the Fermi energy.  
(Hint: Take  $(E - E_F) = kT$ )

(Ans:  $f(E) = 0.269$ )

8. In a solid, consider the energy level lying  $0.01 \text{ eV}$  below Fermi level. What is the probability of this level not being occupied by electron?

(Ans:  $[1 - f(E)] = 0.405$ )

9. Calculate the probabilities for an electronic state to be occupied at  $20^\circ \text{C}$ , if the energy of these states lies  $0.11 \text{ eV}$  above and  $0.11 \text{ eV}$  below the Fermi level

(Ans:  $f(E) = 0.987$ )

10. What is the probability of an electron being thermally excited to the conduction band at Si at 27° C. the band gap is 1.12 eV.

(Ans:  $3.97821 \times 10^{-10}$ )

11. Fermi energy of silver is 5.5 eV. Find out the energy for which the probability of occupancy at 300K is 0.9.

(Ans: 5.44 eV)

12. The Hall coefficient of a specimen is  $3.66 \times 10^{-4} \text{ m}^3 / \text{C}$ , its resistivity is  $8.93 \times 10^{-4} \Omega \text{ m}$ , find  $\mu$  and  $n$

(Ans:  $\mu = 0.041 \text{ m}^2 / \text{Vs}$ ,  $n = 1.7 \times 10^{22} / \text{m}^3$ )

13. In a Hall effect experiment set up a sample of n type Ge has a donor density of  $10^{21} / \text{m}^3$ . Find the Hall voltage developed if the magnetic field used is 0.6 T, current density  $J$  is  $500 \text{ A} / \text{m}^2$  and the thickness of the sample is 5 mm

(Ans: Hall voltage ( $V_H$ ) = 9.375 mV)

14. A Cu strip 2 cm wide and 1 mm thick placed in magnetic field  $B = 1.5 \text{ wb} / \text{m}^2$  a current of 200 Amp, is set up in the strip, calculate the Hall voltage that appeared across the strip ( $R_H = 6 \times 10^{-7} \text{ m}^3 / \text{C}$ ).

(Ans: Hall voltage = 9 mV)

15. A sample of n type Silicon has a donor density of  $10^{20} / \text{m}^3$ . It is used in the Hall effect experiment. If the sample of thickness 4.5 mm is kept in a magnetic field of 0.55 T with current density of  $500 \text{ A} / \text{m}^2$ . Find (i) Hall

voltage developed in it, (ii) Hall coefficient, (iii) Hall angle if mobility of electron is  $0.17 \text{ m}^3/\text{V-s}$ .

(Ans: (i) Hall voltage =  $77.3 \text{ mV}$ )

(ii) Hall coefficient =  $0.0625 \text{ m}^3/\text{C}$

(iii) Hall angle =  $5.36^\circ$ )

16. In a semiconductor with Hall coefficient  $145 \text{ cc/C}$  having width  $2 \text{ cm}$  and thickness  $0.2 \text{ cm}$  with a magnetic field induction of  $2 \text{ T}$  along the smaller dimension, a current of  $150 \text{ mA}$  is calculate the current density and hall voltage.

(Ans:  $J = 3750 \text{ A/m}^2$  and  $V_H = 21.75 \text{ mV}$ )

17. N type Ge sample has donor concentration  $10^{21} \text{ atoms/m}^3$ . What Hall voltage would you expect if current of  $1 \text{ mA}$  and magnetic field  $0.5 \text{ T}$  is applied across  $2 \text{ mm}$  thick sample

(Ans:  $V_H = 1.56 \text{ mV}$ )

